

Day 06 - Mastering Advanced Linux Commands

Objective

To learn advanced Linux commands for file management, process monitoring, system administration, networking, and productivity. These commands are essential for managing Linux servers in Cloud Computing and DevOps environments.

Advanced File & Directory Management

Linux provides powerful commands to create, move, copy, delete, and organize files efficiently.

Search for Files

```
find /path -name "filename"
```

Example:

```
find . -name "*.txt"
```

Searches for all `.txt` files in the current directory and its subdirectories.

Searching File Content

The `grep` command searches for specific words or patterns inside files.

```
grep "keyword" filename
```

Example:

```
grep "error" logs.txt
```

Useful for:

- Searching log files
 - Finding configuration values
 - Debugging applications
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Viewing Running Processes

Every running program in Linux is represented as a process.

Display Running Processes

```
ps
```

Shows currently running processes.

Monitor Processes in Real Time

```
top
```

Displays:

- CPU Usage
 - Memory Usage
 - Running Processes
 - System Load
-

File Permissions

Linux uses permissions to control who can read, write, or execute files.

Change Permissions

```
chmod 755 filename
```

Common permission values:

Permission	Meaning
777	Full access for everyone
755	Owner has full access, others can read and execute
644	Owner can read/write, others can only read

Note: Avoid using `777` unless absolutely necessary, as it grants unrestricted access.

File Ownership

Every file belongs to a user and a group.

Change Owner

```
sudo chown username filename
```

Example:

```
sudo chown arpit notes.txt
```

Disk Usage

Check Disk Space

```
df -h
```

Displays:

- Total disk space
- Used space
- Available space
- Mounted file systems

Check Directory Size

```
du -sh folder-name
```

Displays the total size of a directory in a human-readable format.

Viewing File Contents

Display First Lines

```
head filename
```

Display Last Lines

```
tail filename
```

These commands are especially useful for reading large log files.

Command History

```
history
```

Shows previously executed commands, making it easy to repeat or review past work.

Downloading Files

Using wget

```
wget https://example.com/file.zip
```

Downloads files directly from the internet.

Using curl

```
curl https://example.com
```

Common uses:

- Downloading files
 - Testing APIs
 - Sending HTTP requests
 - Troubleshooting web services
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Network Connectivity

Test Network Connection

```
ping google.com
```

Displays:

- Connectivity status
- Response time
- Packet loss

Useful for checking whether a server is reachable.

Command Chaining

Linux allows multiple commands to be executed in a single line.

Execute Sequentially

```
mkdir project && cd project
```

The second command runs only if the first command succeeds.

Benefits:

- Faster workflow
 - Better automation
 - Cleaner shell scripts
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Commands Practiced Today

```
find  
grep  
ps  
top  
chmod  
chown  
df -h  
du -sh  
head  
tail  
history  
wget  
curl  
ping
```

Hands-On Practice

- Searched files using `find`
- Searched text using `grep`

- Viewed running processes with `ps`
 - Monitored system performance using `top`
 - Changed file permissions using `chmod`
 - Changed ownership using `chown`
 - Checked disk usage
 - Measured directory size
 - Viewed file contents with `head` and `tail`
 - Reviewed command history
 - Downloaded files using `wget` and `curl`
 - Tested network connectivity using `ping`
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Key Learnings

- Linux provides powerful tools for managing files and directories.
 - File permissions and ownership help secure the operating system.
 - System monitoring commands help identify performance issues.
 - Network utilities are essential for troubleshooting servers.
 - Command chaining increases productivity and simplifies repetitive tasks.
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Why This Matters for Cloud & DevOps

Cloud and DevOps engineers spend much of their time working in Linux terminals.

These commands are essential for:

- Managing cloud servers
- Monitoring applications
- Debugging production issues
- Configuring permissions

- Checking storage usage
 - Automating administrative tasks
 - Troubleshooting network connectivity
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Interview Questions

1. What is the difference between `find` and `grep` ?
 2. What does the `top` command display?
 3. What is the purpose of `chmod` ?
 4. How is `chown` different from `chmod` ?
 5. What is the difference between `df -h` and `du -sh` ?
 6. When would you use `head` and `tail` ?
 7. What is the difference between `wget` and `curl` ?
 8. Why is the `ping` command useful?
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Day 06 Summary

Today I explored advanced Linux commands that are commonly used in system administration and DevOps. I learned how to search files, monitor processes, manage permissions, analyze disk usage, test network connectivity, and improve productivity using command chaining. These skills are essential for managing Linux-based cloud infrastructure efficiently.

Status:  Day 06 Completed